Vinod Tiwari

Data Scientist | AI Engineer | AI Architect

vinod.jrc@outlook.com | +91-9559856270 | New Delhi, India | Portfolio | GitHub | Medium | LinkedIn

Professional Summary

Results-driven **Data Scientist, AI Engineer, and AI Architect** with **7+ years of experience** designing, developing, and deploying **production-grade Machine Learning, Deep Learning, NLP, and Generative AI systems**. Deep expertise in **LLM fine-tuning, Retrieval-Augmented Generation (RAG), Agentic AI, multi-agent orchestration (LangGraph, LangChain), and MLOps**.
Proven track record at **Deloitte** and **TCS** delivering end-to-end **AI/ML pipelines** on **AWS, Azure, and Docker/Kubernetes** with measurable business impact — reducing inference costs by 30%+, improving retrieval precision to 92%, and automating workflows saving 20+ hours per week. Skilled in **AI system architecture**, **model optimization**, **cloud-native deployments**, and **cross-functional technical leadership**.
Seeking senior roles as **Data Scientist | AI Engineer | AI Architect** to architect intelligent, scalable AI solutions at enterprise scale.

Technical Skills

Programming & Scripting:	Python, SQL, C, Unix Shell Scripting
AI / ML / Deep Learning:	Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Large Language Models (LLMs), Transformers, BERT, GPT, Model Quantization, Transfer Learning
Generative AI & Agents:	Retrieval-Augmented Generation (RAG), Graph RAG, Agentic AI, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, Prompt Engineering, LLM Evaluation, Guardrails, Model Context Protocol (MCP)
Data Science & Analytics:	NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Feature Engineering, Statistical Modeling, Time Series Forecasting (ARIMA), Demand Forecasting, Sentiment Analysis
Frameworks & MLOps:	TensorFlow, Docker, Kubernetes, CI/CD Pipelines, Model Deployment, Serverless Pipelines, REST API (FastAPI), Streamlit
Cloud & Databases:	AWS (Lambda, S3, Amazon Neptune, SageMaker),PostgreSQL, VectorDB (Pgvector/FAISS),
Tools & Platforms:	Git, Jupyter, VS Code, Salesforce (AI integration), OCR (Tesseract, OpenCV), PyPDF, PDFPlumber
Architecture & Leadership:	AI System Architecture, Microservices Design, Scalable Pipeline Architecture, Technical Leadership, Cross-Functional Collaboration, Agile / Scrum

Professional Experience

Deloitte USI — **Jun 2024 — Present**

Senior AI Engineer | Generative AI & Agentic Systems Architect

Delivered enterprise-scale Agentic AI, Graph RAG, and LLM evaluation systems — improving generation efficiency by up to 70% and retrieval precision to 92% across production deployments.

- **Architected** an **Agentic AI multi-agent CSR system** using **LangGraph, MCP, and AWS**, achieving **90%+ SME accuracy**, improving generation efficiency by **60–70%**, and reducing LLM hallucinations by **35%** via LLM-as-Judge evaluation (precision, recall, faithfulness).
- **Delivered** a **Graph RAG pipeline on AWS** (Amazon Neptune + VectorDB) for vehicle lifecycle tracking, improving retrieval precision to **92%**, relationship accuracy by **40%**, and cutting query latency by **45%**; implemented enterprise **LLM guardrails** (prompt injection, adversarial testing), reducing failures by **40%** and sustaining **99.9% uptime**.
- **Developed** a **Project Intelligence Platform** for managers to estimate client cost and delivery timelines — built custom **manual embedding pipelines (no LLM)** over **PostgreSQL + VectorDB**, enabling **cosine similarity search** to retrieve the **top-10 most relevant past Deloitte projects** by client requirements, improving estimation accuracy and efficiency by **50%**.
- **Engineered** a **Talk-to-Infra conversational agent** for MongoDB diagnostics and a **serverless PDF extraction pipeline** (PyPDF, PDFPlumber, AWS Lambda), processing **10,000+ documents/day** at **95%+ accuracy**, cutting resolution time by **50%** and infrastructure costs by **35%**.
- **Implemented** a **RAG-based AI Chat Plugin for Salesforce ticket resolution**, boosting low-priority auto-resolution by **40%** and reducing agent response time by **30%**.
- **Led** design of a **healthcare value-based care proforma platform** (Linear Regression, Random Forest), improving prediction accuracy by **25%**, reducing evaluation time by **60%**, and optimizing contract costs by **15–20%**.

Tata Consultancy Services (TCS) — **Jan 2019 — May 2024**

Data Scientist | AI/ML Engineer — R&D, Retail & Computer Vision

Built production ML, NLP, and Computer Vision systems for large-scale retail AI — from model quantization and OCR pipelines to LLM POCs and demand forecasting, with an IEEE WACV 2024 research contribution.

- **Optimized** production ML models via **model quantization (AMD framework)**, reducing inference cost by **30%+** while maintaining accuracy within $<$**2% degradation**; designed and fine-tuned **CNN models (InceptionV3, ResNet, EfficientNet-B0)** on **26K+ images**, achieving **98.9% classification accuracy** for retail product categorization.
- **Delivered** end-to-end **OCR and Computer Vision pipelines** (CRAFT, OpenCV, Tesseract), achieving **95%+ text detection accuracy** and boosting OCR accuracy by **30%**; enhanced text similarity using **semantic embeddings (Word2Vec, Transformers)**.
- **Contributed** to a **WACV 2024 (IEEE) publication** on retail taxonomy classification via **Progressive Multi-Task Learning (PMTL)**; resolved hierarchical label inconsistency and error propagation using logit masking, improving fine-grained retrieval efficiency.
- **Implemented LLM POCs (GPT-3, BERT, LangChain)** with scalable REST APIs (Flask, Docker), reducing deployment cycle time by **30%**; developed **NLP sentiment systems** (VADER, TextBlob, Transformers) improving prediction accuracy by **20–25%**.
- **Built demand forecasting models** (Ridge, Lasso, ARIMA) improving sales accuracy by **15–20%** and **ETL pipelines on Snowflake (Azure)** cutting processing time by **30%**, ensuring high data quality at scale.
- **Automated** daily reporting workflows in Python, saving **20+ hours/week**; recognized as **SME (Python, ML)** and awarded TCS Golden Guru Gala and Faculty Awards (2021–2022).

Education

Bachelor of Technology (B.Tech), Uttar Pradesh Technical University — 2014 — 2018
Intermediate (Mathematics), A.I.T. Inter College — 2010 — 2012

Awards & Recognition

TCS Golden Guru Gala Award, Tata Consultancy Services — 2021
Recognized for excellence in mentoring and educating teams on **Machine Learning, Deep Learning**, and **Python programming** with clear, visually compelling instructional methods.

TCS Faculty Award (Subject Matter Expert), Tata Consultancy Services — 2021–2022
Honored twice as **Subject Matter Expert** in Python and Machine Learning; featured in the TCS Yearbook (2021, 2022).

Certifications & Training

AWS Partner: Agentic AI Essentials, Amazon Web Services — 2026
AWS Partner: Generative AI Technical, Amazon Web Services — 2026
Agentic AI Practicum, Deloitte Internal — 2025
Machine Learning Specialization, Stanford University / Coursera (Andrew Ng) — 2022
Remote Sensing and Deep Learning, Indian Space Research Organization (ISRO) — 2021

Projects & Publications

- **AI-Doctor Application (In Progress):** Architecting an **agent-based AI healthcare platform** integrating health report analysis, personalized diet/gym plan generation, and medical recommendation systems using **multi-agent orchestration and LLMs**; presented at **Deloitte USI Hackathon**; currently deploying as a **full-stack production-grade application**.

- **WACV 2024 (IEEE) Research Contribution:** Co-contributed to a published research paper on **retail taxonomy classification using Progressive Multi-Task Learning (PMTL)**, addressing hierarchical label inconsistency in large-scale product datasets.

- **Dancing Robot (MNNIT Allahabad Expo, 2017):** Designed and presented a real-time motion-controlled robot using **C programming and Arduino**, demonstrating embedded systems and hardware-software integration.

- **3D Printer Prototype (2017):** Built a functional 3D printer prototype applying **mechatronics principles**, including hardware design, motion control, and system integration.